# Schottky Power Rectifier, Surface Mount, Fast Soft−Recovery 200 V, 3.0 A

## SMC Power Surface Mount Package

# MBRS3201T3G, NRVBS3201T3G

### Features

- Lower Forward Voltage than any Ultrafast Rectifier: $V_F$ < 0.59 V at 150°C
- Fast Switching Speed: Reverse Recovery Time ($t_{RR}$) < 35 ns
- Soft Recovery Characteristics: Softness Factor ($t_b/t_a$) ≥ 1
- Highly Stable Over Temperature
- NRVB Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC−Q101 Qualified and PPAP Capable*
- These Devices are Pb−Free, Halogen Free/BFR Free and are RoHS Compliant

### Benefits

- Significantly Reduced EMI
- Eliminates the Need of Snubber Circuits
- Low Switching and Heat Losses
- Improved Thermal Management

### Applications

- Engine and Convenience Control Systems
- Motor Controls
- Battery Chargers and Switching Power Supplies

### Mechanical Characteristics

- Small Compact Surface Mount Package with J-Bend Leads
- Rectangular Package for Automated Handling
- Weight: 217 mg (Approximately)
- Finish: All External Surfaces Corrosion Resistant and Terminal Leads are Readily Solderable
- ESD Ratings:
  - Machine Model = A
  - Human Body Model = 1C
- Lead and Mounting Surface Temperature for Soldering Purposes: 260°C Maximum for 10 Seconds
- Polarity: Polarity Band on Plastic Body Indicates Cathode Lead

## SCHOTTKY RECTIFIER
## 3 AMPS, 200 VOLTS



**SMC 2−LEAD**
**CASE 403AC**



### MARKING DIAGRAM




| | |
|---|---|
| B321 | = Specific Device Code |
| A | = Assembly Location** |
| Y | = Year |
| WW | = Work Week |
| ▪ | = Pb−Free Package |

(Note: Microdot may be in either location)

**The Assembly Location code (A) is front side optional. In cases where the Assembly Location is stamped in the package, the front side assembly code may be blank.

### ORDERING INFORMATION

| Device | Package | Shipping† |
|---|---|---|
| MBRS3201T3G | SMC 2−LEAD (Pb−Free) | 2500 / Tape & Reel |
| NRVBS3201T3G* | SMC 2−LEAD (Pb−Free) | 2500 / Tape & Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

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## MAXIMUM RATINGS

| Characteristic | Symbol | Value | Unit |
|---|---|---|---|
| Peak Repetitive Reverse Voltage<br>Working Peak Reverse Voltage<br>DC Blocking Voltage | $V_{RRM}$<br>$V_{RWM}$<br>$V_R$ | 200 | V |
| Average Rectified Forward Current (Rated $V_R$, $T_C$ = 70°C) | $I_{F(AV)}$ | 3 | A |
| Nonrepetitive Peak Surge Current (83 ms, half−sine) | $I_{FSM}$ | 100 | A |
| Operating Junction Temperature | $T_J$ | −55 to +150 | °C |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

## THERMAL CHARACTERISTICS

| Characteristic | Symbol | Value | Unit |
|---|---|---|---|
| Thermal Resistance, Junction−to−Lead | $R_{\theta JL}$ | 12 | °C/W |
| Thermal Resistance, Junction−to−Ambient | $R_{\theta JA}$ | 60 | °C/W |

## ELECTRICAL CHARACTERISTICS

| Characteristic | Symbol | Value | Unit |
|---|---|---|---|
| Maximum Instantaneous Forward Voltage<br>($I_F$ = 3 A, $T_J$ = 25°C)<br>($I_F$ = 3 A, $T_J$ = 150°C) | $V_F$ | <br>0.84<br>0.59 | V |
| Maximum Instantaneous Reverse Current (Rated $V_R$)<br>(Rated DC Voltage, $T_J$ = 25°C)<br>(Rated DC Voltage, $T_J$ = 150°C) | $I_R$ | <br>1.0<br>5.0 | <br>mA<br>mA |
| Maximum Reverse Recovery Time<br>($I_F$ = 1 A, di/dt = 100 A/us, $V_R$ = 30 V) | $t_{rr}$ | <br>35 | ns |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

## TYPICAL ELECTRICAL CHARACTERISTICS

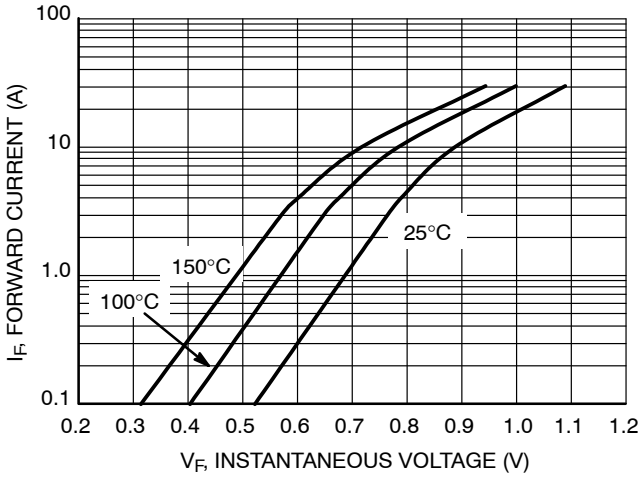



**Figure 1. Typical Forward Voltage**

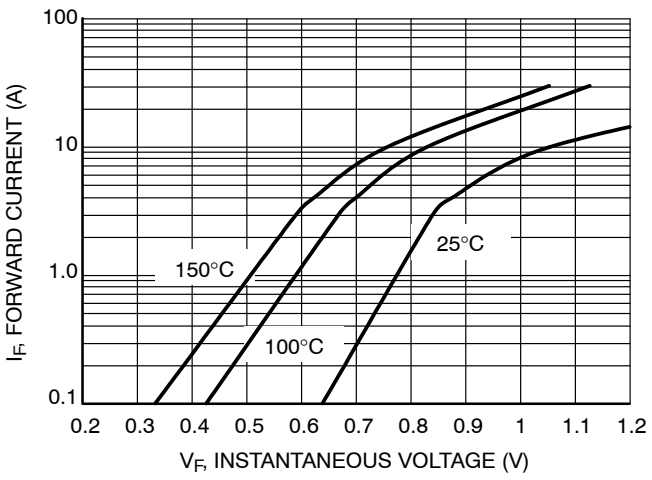



**Figure 2. Maximum Forward Voltage**



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## TYPICAL ELECTRICAL CHARACTERISTICS (CONTINUED)

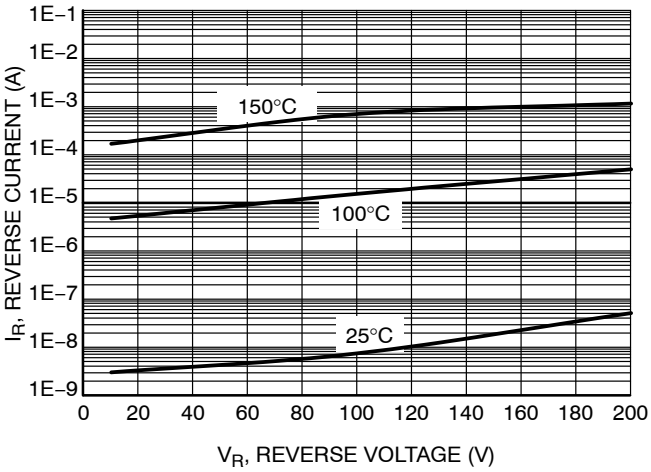



**Figure 3. Typical Reverse Current**

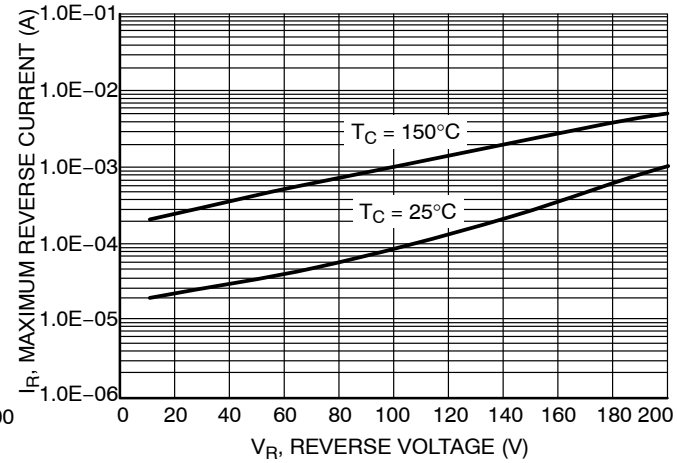



**Figure 4. Maximum Reverse Current**

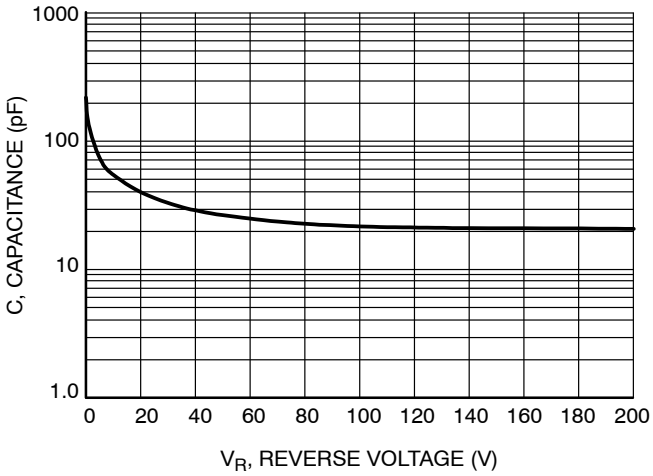



**Figure 5. Typical Capacitance**

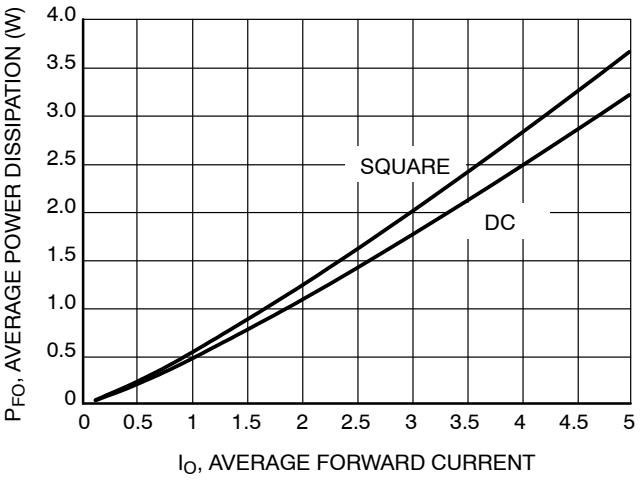



**Figure 6. Power Dissipation**

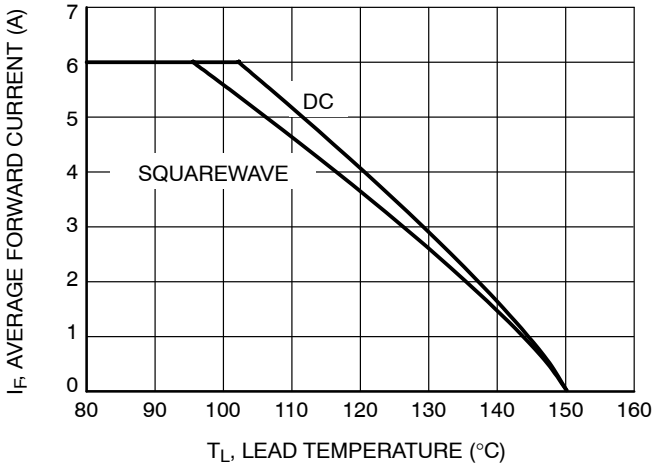



**Figure 7. Derating Curve**



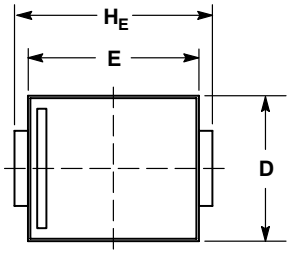
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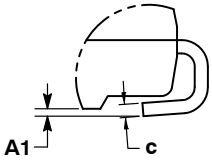
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**SMC 2−LEAD**
CASE 403AC
ISSUE B

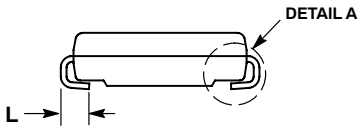
DATE 27 JUL 2017



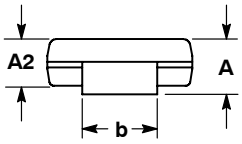

**TOP VIEW**




**DETAIL A**




**SIDE VIEW**

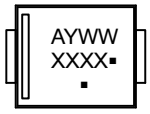



**END VIEW**

NOTES:
1. DIMENSIONING AND TOLERANCING PER ANME Y14.5M, 1994.
2. CONTROLLING DIMENSION: INCHES.
3. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH. MOLD FLASH SHALL NOT EXCEED 0.254mm PER SIDE.
4. DIMENSIONS D AND E TO BE DETERMINED AT DATUM H.
5. DIMENSION b SHALL BE MEASURED WITHIN THE AREA DETERMINED BY DIMENSION L.

| | MILLIMETERS | | INCHES | |
|---|---|---|---|---|
| DIM | MIN | MAX | MIN | MAX |
| A | 1.95 | 2.61 | 0.077 | 0.103 |
| A1 | 0.05 | 0.20 | 0.002 | 0.008 |
| A2 | 1.90 | 2.41 | 0.075 | 0.095 |
| b | 2.90 | 3.20 | 0.114 | 0.126 |
| c | 0.15 | 0.41 | 0.006 | 0.016 |
| D | 5.55 | 6.25 | 0.219 | 0.246 |
| E | 6.60 | 7.15 | 0.260 | 0.281 |
| $H_E$ | 7.75 | 8.15 | 0.305 | 0.321 |
| L | 0.75 | 1.60 | 0.030 | 0.063 |

## GENERIC MARKING DIAGRAM*

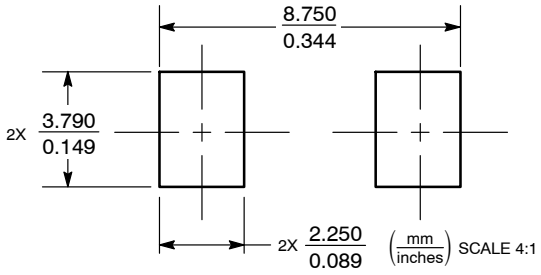



XXXX = Specific Device Code
A = Assembly Location
Y = Year
WW = Work Week
▪ = Pb−Free Package

(Note: Microdot may be in either location)

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

## RECOMMENDED SOLDERING FOOTPRINT*




*For additional information on our Pb−Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

| | | | |
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